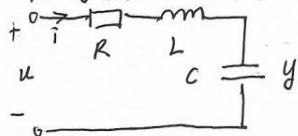


第一周作业

1. 写出RLC串联电路状态空间模型



$$x_1 = y = u_c$$

$$x_2 = \frac{dy}{dt}$$

$$u(t) = u_R + u_L + u_C$$

$$i = C \frac{du_C}{dt} = C \frac{dy}{dt}$$

$$u = CR \frac{du_C}{dt} + LC \frac{d^2 u_C}{dt^2} + u_C$$

$$= CR x_2 + LC \dot{x}_2 + x_1$$

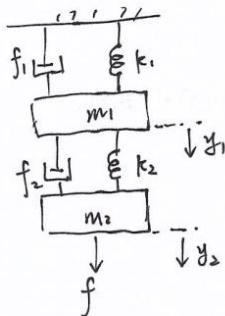
$$\dot{x}_2 = -\frac{1}{LC} x_1 - \frac{R}{L} x_2 + \frac{1}{LC} u$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{1}{LC} x_1 - \frac{R}{L} x_2 + \frac{1}{LC} u \end{cases}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

3.3. 对下图所示的机械系统, 试列以拉力 F 为输入, 质量块 m_1 和 m_2 的位移 y_1 和 y_2 为输出的状态空间表达式。



解: 由 $\sum F = ma$ 可列方程如下:

$$f - f_2(y_2 - y_1) - k_2(y_2 - y_1) = m_2 \ddot{y}_2$$

$$f_2(\dot{y}_2 - \dot{y}_1) + k_2(y_2 - y_1) - f_1 \dot{y}_1 - k_1 y_1 = m_1 \ddot{y}_1$$

$$\text{可得: } \begin{cases} m_1 \ddot{y}_1 + f_1 \dot{y}_1 + k_1 y_1 = f_2(\dot{y}_2 - \dot{y}_1) + k_2(y_2 - y_1) \\ m_2 \ddot{y}_2 + f_2(\dot{y}_2 - \dot{y}_1) + k_2(y_2 - y_1) = f \end{cases}$$

∴ 该系统有 4 个 独立的 储能元件。

∴ 四个状态变量, 令 $x_1 = y_1$, $x_2 = y_2$, $\dot{x}_3 = \dot{y}_1$, $\dot{x}_4 = \dot{y}_2$

$$\text{可得: } \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_1 + k_2}{m_1} & \frac{k_2}{m_1} & -\frac{f_1 + f_2}{m_1} & \frac{f_2}{m_1} \\ \frac{k_2}{m_2} & -\frac{k_2}{m_2} & \frac{f_2}{m_2} & -\frac{f_2}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} f$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

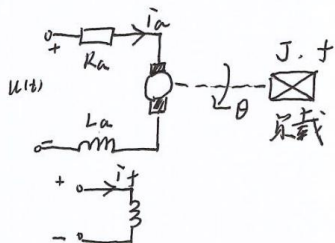
1. 有同学会考虑重力, 本题忽略重力。

3. 一些正负号错误。

2. k_2 与 f_2 产生的力为 $k_2(y_2 - y_1)$ 和 $f_2(\dot{y}_2 - \dot{y}_1)$, 不是 $k_2 y_2$ 和 $f_2 \dot{y}_2$

3.4 图为带载直流电机工作原理图。工作期间必须时刻保持恒定。带载电机转动惯量为 J ，反电动势系数 k_b ，带载电机黏性阻尼系数为 τ 。输入变量为电枢电压 $u(t)$ ，输出为电机轴角位移 $\theta(t)$ 。试分别选状态变量为

(1) $x^T = [\theta, \dot{\theta}, i_a]$ (2) $x^T = [\theta, \dot{\theta}, \ddot{\theta}]$ 列出系统状态空间模型。



解：机械部分： $M = J \frac{dw}{dt} + \tau w$

电磁部分： $u = R_a i_a + L_a \frac{di_a}{dt} + e_a$

机电耦合： $e_a = k_b w$

$m = k_d i_a$

(1) $x^T = [\theta, \dot{\theta}, i_a]$

$$\frac{dw}{dt} = \ddot{\theta} \quad w = \dot{\theta}$$

$$k_d i_a = J \ddot{\theta} + \tau \dot{\theta}$$

$$u = R_a i_a + L_a \dot{i}_a + k_b \dot{\theta}$$

整理可得

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{\tau}{J} x_2 + \frac{k_d}{J} x_3 \\ \dot{x}_3 = -\frac{k_b}{L_a} x_2 - \frac{R_a}{L_a} x_3 + \frac{1}{L_a} u \end{cases}$$

$$y = \theta = [1 \ 0 \ 0] x$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\tau}{J} & \frac{k_d}{J} \\ 0 & -\frac{k_b}{L_a} & -\frac{R_a}{L_a} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_a} \end{bmatrix} u$$

$$y = [1 \ 0 \ 0] x$$

(2) $x^T = [\theta, \dot{\theta}, \ddot{\theta}]$

$$k_d i_a = J \ddot{\theta} + \tau \dot{\theta}$$

$$u = R_a i_a + L_a \dot{i}_a + k_b \dot{\theta}$$

$$\dot{i}_a = \frac{J}{L_a} \ddot{\theta} + \frac{\tau}{L_a} \dot{\theta}$$

$$\frac{di_a}{dt} = \frac{J}{L_a} \ddot{\theta} + \frac{\tau}{L_a} \dot{\theta}$$

代入可得

$$u = \frac{J R_a}{k_d} \ddot{\theta} + \frac{\tau R_a}{k_d} \dot{\theta} + \frac{J L_a}{k_d} \ddot{\theta} + \frac{\tau L_a}{k_d} \dot{\theta} + k_b \dot{\theta}$$

整理可得

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = -\frac{\tau R_a + k_d k_b}{J L_a} x_2 - \frac{J R_a + \tau L_a}{J L_a} x_3 + \frac{k_d}{J L_a} u \end{cases}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -\frac{\tau R_a + k_d k_b}{J L_a} & -\frac{J R_a + \tau L_a}{J L_a} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \frac{k_d}{J L_a} \end{bmatrix} u$$

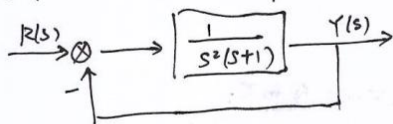
$$y = [1 \ 0 \ 0] x$$

3.5. $\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{cases}$ 求该系统的传递函数.

解: $G(s) = C(sI - A)^{-1}B + D = \frac{s}{s^2 + 5s + 7}$.

计算错误.

3.6. 系统框图如图所示, 试选取合适的状态变量, 列出该系统的状态空间模型.



解: 闭环传递函数 $\frac{Y(s)}{R(s)} = \frac{\frac{1}{s^2(s+1)}}{1 + \frac{1}{s^2(s+1)}} = \frac{1}{s^3 + s^2 + 1}$. 即: $(s^3 + s^2 + 1)Y(s) = R(s)$.

由拉氏反变换可以得到: $y^{(3)}(t) + y^{(2)}(t) + y(t) = r(t)$

令 $x_1(t) = y(t)$, $x_2(t) = y'(t)$, $x_3(t) = y''(t)$. 则:

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r \\ y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \end{cases}$$

1. 闭环传递公式没记对. 2. 系统阶次判断错误, 写为四阶系统.

3.7

$\ddot{y} + 3\dot{y} + 2y = 2\dot{u} + u$
求系统状态空间模型.

拉氏变换:

$$s^3 Y(s) + 3s^2 Y(s) + 2Y(s) = 2sU(s) + U(s)$$

$$= 2sU(s) + U(s)$$

$$\text{即 } Y(s) = \frac{2s+1}{s^3+3s^2+2s} U(s)$$

$$\text{设 } \tilde{Y}(s) = \frac{1}{s^3+3s^2+2s} U(s)$$

$$\begin{cases} s^3 \tilde{Y}(s) + 3s^2 \tilde{Y}(s) + 2s \tilde{Y}(s) = U(s) \\ Y(s) = 2s \tilde{Y}(s) + \tilde{Y}(s) \end{cases}$$

$$\text{令 } x_1(t) = \tilde{y}(t) \quad x_2(t) = \dot{\tilde{y}}(t) \quad x_3(t) = \ddot{\tilde{y}}(t)$$

$$\text{则 } \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = -2x_2 - 3x_3 + u \end{cases}$$

$$y = x_1 + 2x_2$$

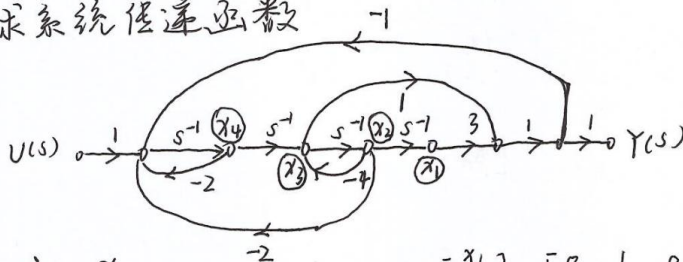
$$\text{即 } \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 2 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

3.8 题 3.8 图所示为系统状态图

(1) 确定状态变量并写出系统状态方程

(2) 求系统传递函数



$$(1) \quad \dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3 - 4x_2$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = u - 2x_2 - 2x_4 - y$$

$$y = 3x_1 + x_3 - 4x_2$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -3 & 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [3 \quad -4 \quad 1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$(2) \quad G(s) = C(sI - A)^{-1}B + D = \frac{s^2 + 3}{s^4 + 6s^3 + 9s^2 + 3}$$

3.9 设 n 阶线性系统传递函数为

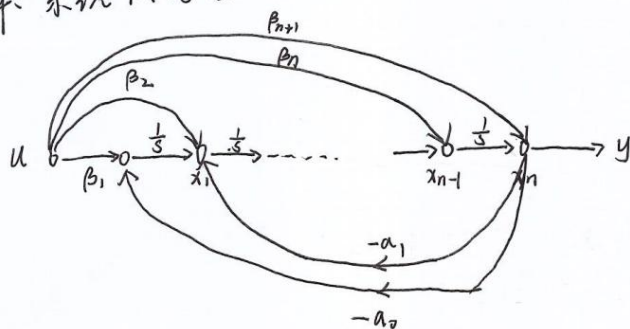
$$G(s) = \frac{b_n s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

试将选择合适的状态变换转化为如下形式状态空间模型:

$$\dot{x} = \begin{bmatrix} 0 & \dots & -a_1 \\ 1 & \dots & -a_2 \\ & \ddots & \vdots \\ & & 1 & -a_{n-1} \end{bmatrix} x + \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix} u$$

并确定 $\beta_1, \dots, \beta_n$ 值。绘图系统状态图 (上述状态空间模型为能观规范型)

解: 系统状态图为



根据题意 $D=0$, 即 $b_n=0$

$$\dot{x}_1 = -a_0 x_n + \beta_1 u$$

$$\dot{x}_2 = x_1 - a_1 x_n + \beta_2 u$$

$$\vdots$$

$$\dot{x}_n = x_{n-1} - a_{n-1} x_n + \beta_n u$$

$$y = x_n$$

$$\therefore \dot{x} = \begin{bmatrix} 0 & \dots & -a_0 \\ 1 & \dots & -a_1 \\ & \ddots & \vdots \\ & & 1 & -a_{n-1} \end{bmatrix} x + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} u$$

$$y = [0 \quad \dots \quad 0 \quad 1] x$$

$$\beta_1 = b_0, \quad \beta_2 = b_1, \quad \dots, \quad \beta_n = b_{n-1}$$

$$\text{当 } b_n \neq 0 \text{ 时} \quad \beta_i = b_{i-1} - a_{i-1} b_n$$

$$y = [0, \dots, 0, 1] x + b_n u$$

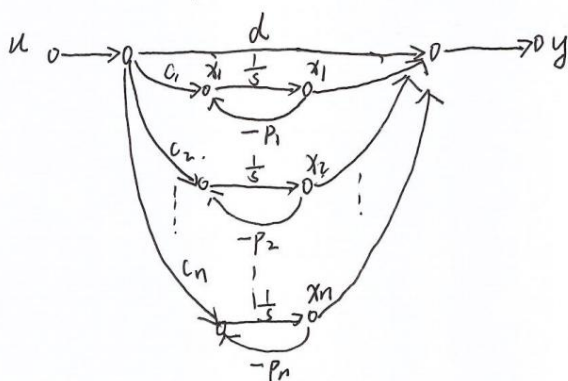
3.10 当系统各极点互异时, 其传递函数表示为

$$G(s) = d + \sum_{i=1}^n \frac{c_i}{s + \beta_i}$$

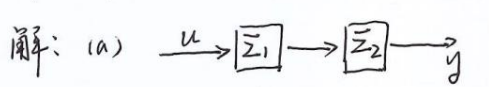
试说明为一种对角线解耦规范型为

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{cccc|c} -\beta_1 & \cdots & 0 & & c_1 \\ & \ddots & & & \vdots \\ & & -\beta_n & & c_n \\ \hline 1 & \cdots & 1 & & d \end{array} \right]$$

解: 系统流图为



3.11 两个子系统 $\bar{z}_1(A_1 B_1 C_1)$ $\bar{z}_2(A_2 B_2 C_2)$ 分别进行串, 并, 反馈连接, 试求三种连接后系统的 A, B, C 矩阵



$$u = u_1 \quad u_2 = y_1 \quad y_2 = y$$

$$\dot{x}_1 = A_1 x_1 + B_1 u$$

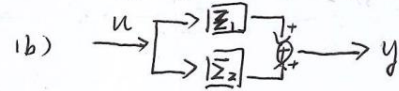
$$\dot{x}_2 = A_2 x_2 + B_2 y_1$$

$$= A_2 x_2 + B_2 C_1 x_1$$

$$y = y_2 = C_2 x_2$$

$$\therefore A = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix} \quad B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & C_2 \end{bmatrix}$$



$$u = u_1 = u_2$$

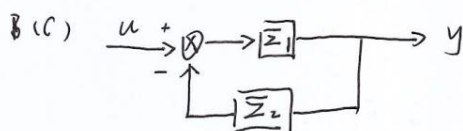
$$y = y_1 + y_2$$

$$\dot{x}_1 = A_1 x_1 + B_1 u$$

$$\dot{x}_2 = A_2 x_2 + B_2 u$$

$$\therefore A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

$$C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}$$



$$u_1 = u - y_2 \quad y_1 = y = u_2$$

$$\dot{x}_1 = A_1 x_1 + B_1 u - B_1 C_2 x_2$$

$$\dot{x}_2 = A_2 x_2 + B_2 C_1 x_1$$

$$y = C_1 x_1$$

$$\therefore A = \begin{bmatrix} A_1 & -B_1 C_2 \\ B_2 C_1 & A_2 \end{bmatrix} \quad B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} C_1 & 0 \end{bmatrix}$$

3.12 三阶线性系统的系统矩阵为 $A = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

分别应用拉普拉斯变换法和凯莱-哈密顿定理计算系统的状态转移矩阵 $\varphi(t)$

① Laplace 变换: $(sI - A)^{-1} = \begin{pmatrix} s & 0 & 2 \\ 0 & s-1 & 0 \\ -1 & 0 & s-3 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{-2s}{s^3-3s^2+2s} & 0 & \frac{2}{s^3-3s^2+2s} \\ 0 & \frac{1}{s-1} & 0 \\ \frac{s}{s^3-3s^2+2s} & 0 & \frac{s}{s^3-3s^2+2s} \end{pmatrix}$

$\varphi(t) = \mathcal{L}^{-1}[(sI - A)^{-1}] = \begin{pmatrix} 2e^t - e^{2t} & 0 & 2e^t - 2e^{2t} \\ 0 & e^t & 0 \\ 2e^t - e^t & 0 & 2e^t - e^t \end{pmatrix}$

② 凯莱-哈密顿法:

特征值: $\lambda_1 = \lambda_2 = 1, \lambda_3 = 2$

$\begin{pmatrix} \alpha_0(t) \\ \alpha_1(t) \\ \alpha_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}^{-1} \begin{pmatrix} te^t \\ e^t \\ e^{2t} \end{pmatrix} = \begin{pmatrix} e^{2t} - 2te^t \\ 3te^t - 2e^{2t} \\ e^{2t} - (t+1)e^t \end{pmatrix}$

$\varphi(t) = \alpha_0(t)I + \alpha_1(t)A + \alpha_2(t)A^2$

* 结果同上

3.12 的问题都在于凯莱-哈密顿定理的计算上, 多数为计算错误.

3.13. 线性定常系统的状态方程为

$\dot{x}(t) = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} x(t) + \begin{pmatrix} 2 \\ 0 \end{pmatrix} u(t), \text{ 初始状态 } x_0 = (0, 1)^T$

(1) 若输入量为 $u(t) = e^{-t} (t \geq 0)$, 试计算状态方程的零输入响应和零状态响应.

(2) 分别求取 $u(t) = 1(t)$ 和 $\delta(t)$ 时, 状态方程的解.

(1) $\varphi(t) = \begin{pmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{pmatrix}$

零输入响应: $y_1 = C x_1(t) = C \varphi(t) x(0) = \begin{pmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{pmatrix}$

零状态响应 $y_2 = C x_2(t) = C \int_0^t \varphi(t-\tau) B u(\tau) d\tau = C \begin{pmatrix} 4te^{-t} - 2e^{-t} + 2e^{-2t} \\ -4te^{-t} + 4e^{-t} - 4e^{-2t} \end{pmatrix}$

3.13 (2) 当 $u(t)=1(t)$ 时.

$$\begin{aligned} x_3(t) &= \varphi(t) x(0) + \int_0^t \varphi(t-\tau) B u(\tau) d\tau \\ &= \begin{pmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{pmatrix} + \int_0^t \begin{pmatrix} 4e^{-(t-\tau)} - 2e^{-2(t-\tau)} \\ -4e^{-(t-\tau)} + 4e^{-2(t-\tau)} \end{pmatrix} d\tau = \begin{pmatrix} -3e^{-t} + 3 \\ 3e^{-t} - 2 \end{pmatrix} \end{aligned}$$

当 $u(t)=\delta(t)$ 时

$$\begin{aligned} x_4(t) &= \varphi(t) x(0) + \int_0^t \varphi(t-\tau) B u(\tau) d\tau \\ &= \varphi(t) x(0) + \varphi(t) B = \begin{pmatrix} 5e^{-t} - 3e^{-2t} \\ -5e^{-t} + 6e^{-2t} \end{pmatrix} \end{aligned}$$

3.13 问题比较多: ① $\varphi(t)$ 求不对.

② 零输入“响应”和零状态“响应”要求 y (书 P106)

很多人只求了状态 x

③ 阶跃响应 $u(t)$ 和冲激响应 $\delta(t) \Rightarrow \int f(t)\delta(t)dt = f(0)$, 很多人不知道.

④ 求 $\int_0^t \varphi(t-\tau) B u(\tau) d\tau$ 时将积分内 $u(\tau)$ 误认为是 $u(t)$ 从而算错.

3.14. 已知矩阵 $\varphi(t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sin t & \cos t \\ 0 & -\cos t & \sin t \end{pmatrix}$. 试问, 它可能是某个系统的状态转移矩阵吗?

$$\varphi(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \neq I \quad \therefore \text{不可能.}$$

3.14 没什么问题. 做错的人不知道 $\varphi(0) = e^{A \cdot 0} = I$ (书 P104)

3.15. 设系统的状态方程 $\dot{x} = Ax$, 当 $x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ 时, 状态响应为 $x(t) = \begin{pmatrix} 2e^{-t} - e^{-3t} \\ -2e^{-t} + 3e^{-3t} \end{pmatrix}$
 当 $x(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ 时, $x(t) = \begin{pmatrix} \frac{3}{2}e^{-t} - \frac{3}{2}e^{-3t} \\ -\frac{3}{2}e^{-t} + \frac{9}{2}e^{-3t} \end{pmatrix}$. 求该系统的矩阵 A 和状态转移矩阵.

$x(t) = \varphi(t)x(0) \Rightarrow$ 联立两个初始状态的状态很容易能解出

$$\varphi(t) = \begin{pmatrix} \frac{3}{2}e^{-t} - \frac{3}{2}e^{-3t} & \frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t} \\ -\frac{3}{2}e^{-t} + \frac{9}{2}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t} \end{pmatrix}$$

~~$\varphi'(t) = A\varphi(t)$~~ $\varphi'(t) = \begin{pmatrix} -\frac{3}{2}e^{-t} + \frac{3}{2}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t} \\ \frac{3}{2}e^{-t} - \frac{9}{2}e^{-3t} & \frac{1}{2}e^{-t} - \frac{3}{2}e^{-3t} \end{pmatrix}$

$$\varphi'(t) = A\varphi(t) \Rightarrow \varphi'(0) = A\underbrace{\varphi(0)}_I \Rightarrow A = \underbrace{\varphi'(0)}_I \cdot \underbrace{\varphi(0)^{-1}}_I = \begin{pmatrix} 0 & 1 \\ -3 & -4 \end{pmatrix}$$

3.15 出错的不多, 错都错在: ① $\varphi(t)$ 求错

② 大部分人求 $\varphi(t)$, 都利用 $(sI - A)^{-1} = \mathcal{L}^{-1}[\varphi(t)]$ 来求 A .

计算量变大导致出错.

可以利用状态转移矩阵的性质: $\varphi'(t) = A\varphi(t)$, $\varphi(0) = A_0$ 来减少计算量.

3.16. 一个离散时间系统的脉冲传递函数为

$$G(z) = \frac{4z^3 - 12z^2 + 13z - 7}{(z-1)(z-2)}$$

试建立系统的状态空间模型。

$$\text{解: } G(z) = 4 + \frac{4z^2 - 7z + 1}{z^2 - 4z + 2}$$

$$\text{令 } X_1(z) = \frac{1}{z^2 - 4z + 2} u(z) \Rightarrow \begin{cases} x_1(k+1) = x_2(k) \\ x_2(k+1) = x_3(k) \\ x_3(k) = 2x_1(k) - 5x_2(k) + 4x_3(k) + u(k) \end{cases}$$

$$y(k) = 4x_3(k) - 7x_2(k) + x_1(k) + 4$$

$$\begin{cases} x_1(k+1) = Gx(k) + Hu(k) \\ y(k) = Cx(k) + Du(k) \end{cases} \Rightarrow G = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix} \quad H = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C = [1, -7, 4], \quad D = 4$$

3.16S217的准确率提高。

个人上错在:

① C矩阵写错, 也就是不会写输出。

② 状态图写错, 要见P117.

3.17 对于通过差分方程

$$y(k+2) + 5y(k+1) + 3y(k) = u(k)$$

$$y(0) = y(1) = 0$$

描述的离散时间系统, 试建立系统的状态空间模型, 并画出状态图。

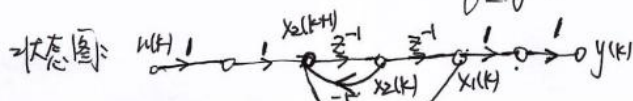
$$\text{传递函数: } G(z) = \frac{1}{z^2 + 5z + 3}$$

$$\text{令 } X_1(z) = \frac{1}{z^2 + 5z + 3} u(z) \Rightarrow \begin{cases} x_1(k+1) = x_2(k) \\ x_2(k+1) = -3x_1(k) - 5x_2(k) + u(k) \end{cases}$$

$$y(k) = x_1(k)$$

$$\Rightarrow \begin{cases} x_1(k+1) = Gx(k) + Hu(k) \\ y(k) = Cx(k) + Du(k) \end{cases} \Rightarrow G = \begin{pmatrix} 0 & 1 \\ -3 & -5 \end{pmatrix} \quad H = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad C = (1, 0)$$

$$D = 0$$



3.18. 线性定常离散时间系统的外态方程为:

$$X(k+1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2 \\ -1 & 1 & 0 \end{pmatrix} X(k) + \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} u(k)$$

初始状态 $X(0) = (1, 2, 3)^T$, 控制序列为 $u(0)=3, u(1)=-6, u(2)=2$,

求 $k=3$ 时离散时间系统的状态 $X(3)$ 。

解: $k=0, X(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2 \\ -1 & 1 & 0 \end{pmatrix} X(0) + \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} u(0) = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$

$k=1, X(2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2 \\ -1 & 1 & 0 \end{pmatrix} X(1) + \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} u(1) = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}$

$k=2, X(3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2 \\ -1 & 1 & 0 \end{pmatrix} X(2) + \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} u(2) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

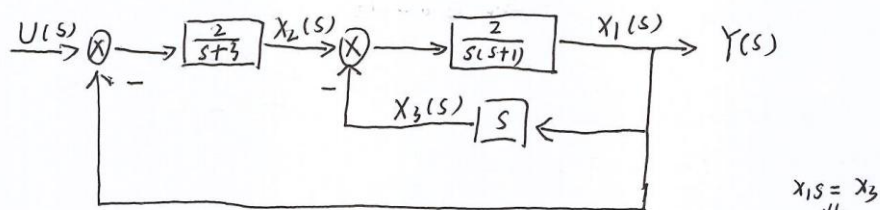
3.18. 错的地方有以下2个

① 计算错误

② 有的同学求状态转移矩阵求不来导致放弃计算。

此题不需要求解状态转移矩阵。

7.4 求图所示系统的状态方程,并判断解控解观性.



解:由图可得

$$[U(s) - X_1(s)] \frac{2}{s+3} = X_2(s)$$

$$X_1(s) s = X_3(s)$$

$$[X_2(s) - X_3(s)] \cdot \frac{2}{s(s+1)} = X_1(s)$$

$$\begin{aligned} X_1 s &= X_3 \\ \downarrow \\ 2(X_2 - X_3) &= s(s+1)X_1 \\ &= (s+1)X_3 \end{aligned}$$

$$\Rightarrow \begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = -2x_1 - 3x_2 - 2u \\ \dot{x}_3 = 2x_2 - 3x_3 \end{cases}$$

$$\dot{X} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} X + \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} u$$

$$\dot{X} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} X + \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} u$$

$$Y = [1 \ 0 \ 0] X$$

$$\text{rank} [B \ AB \ A^2B] = 3$$

完全可控

$$\text{rank} [B \ AB \ A^2B] = 3$$

完全可控

$$\text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = 3$$

完全解观

7.8 判断 (A, B, C) 解控解观性

1) $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = [1 \ 1]$

$$AB = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{rank} [B \ AB] = 1 \quad \text{不完全解控}$$

$$CA = [1 \ 2] \quad \text{rank} \begin{bmatrix} C \\ CA \end{bmatrix} = 2 \quad \text{完全解观}$$

2) $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$$\text{rank} [B \ AB \ A^2B] = \begin{bmatrix} 1 & 0 & -1 & 1 & 1 & -2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 1 & 0 & 2 & 0 & 4 & 0 \end{bmatrix} = 3 \quad \text{完全解控}$$

$$\text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \text{rank} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ -2 & 1 & 0 \\ 0 & -1 & 2 \\ 2 & -3 & 0 \\ 0 & 1 & 4 \end{bmatrix} = 3 \quad \text{完全解观}$$

7.8 (3)

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = [1 \quad 0 \quad 1]$$

解:

$$\text{rank} [B \quad AB \quad A^2B] = \text{rank} \begin{bmatrix} 0 & 1 & 12 \\ 0 & 2 & 8 \\ 1 & 1 & 3 \end{bmatrix} = 3 \quad \text{完全能控}$$

$$\text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \text{rank} \begin{bmatrix} 1 & 0 & 1 \\ 3 & 5 & 2 \\ 9 & 29 & 15 \end{bmatrix} = 3 \quad \text{完全能观}$$

$$14) \quad A = \begin{bmatrix} -2 & 2 & 1 \\ -1 & 2 & 0 \\ 3 & 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = [0 \quad 4 \quad 1]$$

$$\text{rank} [B \quad AB \quad A^2B] = \text{rank} \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & -1 \\ 1 & 1 & 4 \end{bmatrix} = 3 \quad \text{完全能控}$$

$$\text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 1 \\ 0 & 0 & 1 \\ -1 & 9 & 1 \\ 3 & 1 & 1 \\ -4 & 17 & 0 \\ -4 & 9 & 4 \end{bmatrix} = 3 \quad \text{完全能观}$$

7.9. 考虑系统 $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u, y = [G \ C_2 \ C_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$

试问: 除 $G=C_2=C_3=0$ 外, G, C_2, C_3 取何值时系统不能观?

解: 特征值判据: A 的特征值 $\lambda_1=-1, \lambda_2=-2, \lambda_3=-3$. 对 $\forall$ 特征值 $\lambda_i (i=1, 2, 3)$.

当 $\text{rank} \begin{bmatrix} C \\ \lambda_i I - A \end{bmatrix} = 3$ 时, 系统完全能观. $\therefore$ 当有 $\text{rank} \begin{bmatrix} C \\ \lambda_i I - A \end{bmatrix} \leq 2$ 时, 系统不能观.

① $\lambda_1=-1$: $\begin{bmatrix} C \\ -I-A \end{bmatrix} = \begin{bmatrix} G & C_2 & C_3 \\ -1 & -1 & 0 \\ 0 & -1 & -1 \\ 6 & 11 & 5 \end{bmatrix} \xrightarrow{\text{列变换}} \begin{bmatrix} G & C_2-G-C_3 & C_3 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \\ 6 & 0 & 5 \end{bmatrix}$

$\therefore C_2-G-C_3=0$ 时, $\text{rank} \begin{bmatrix} C \\ \lambda_1 I - A \end{bmatrix} = 2$. 系统不能观.

② $\lambda_2=-2$, ③ $\lambda_3=-3$ 同理可得: $G+9C_3-3C_2=0$ 或 $G-2C_2+4C_3=0$ 时, 不能观.

7.8 主要为一些计算错误. 一些同学不能正确处理行数大于1的 C 矩阵形成的 Q_0 ,

对每一行 分开计算了两个 Q_0 分别判断根, 导致错误.

7.9 错误分为如下两种:

① 计算 Q_0 行列式 $=0$ 的条件从而得到 G, C_2, C_3 的条件. 该方法可以求出解但计算量大. 且最后得到的 $|Q_0|$ 还要因式分解. 不适合手动计算场合.

② 得到 $\text{rank} \begin{bmatrix} C \\ \lambda_i I - A \end{bmatrix}$ 表达式后, 不能正确得到其秩 $=2$ 的必要条件, 而是得到了某两行成比例这一充分不必要条件.

7.10. 设系统传递函数为 $G(s) = \frac{s+a}{s^3+7s^2+14s+8}$, 分析当 a 为多大时, 系统将变为不完全能控或不完全能观.

解: $s^3+7s^2+14s+8 = (s+1)(s+2)(s+4)$.

当 $G(s)$ 存在零极点抵消时, 系统不完全能控或不完全能观.

$\therefore a=1$ 或 2 或 4 时.

注: 本题出错的同学少.